Medical Terminology — BBAHM 201

Complete Summary Notes | All 5 Modules Covered

Subject: Medical Terminology | **Code:** BBAHM 201 (Major) | **Credits:** 5 (4T+1P)

University: Maulana Abul Kalam Azad University of Technology, West Bengal (MAKAUT)

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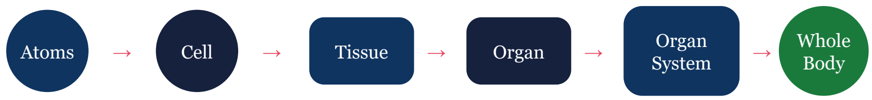
MODULE 1

Basics of Human Body, Body Cavities, Body Planes & Basic Examination

1.1 Levels of Organization of the Human Body

In easy words: Our body is built step by step, like building blocks. Small parts join to make bigger parts, until the whole body is formed.



Picture 1.1 — Levels of the body, from tiny atoms to the whole body, shown small to big

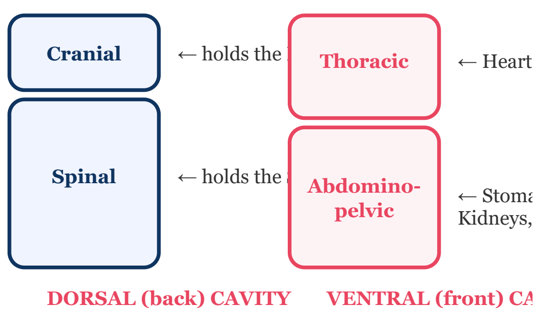
Level	Description	Example
Chemical	Atoms and molecules	Water (H_2O), DNA, proteins
Cellular	Basic structural unit of life	Neuron, Red Blood Cell
Tissue	Group of similar cells performing same function	Muscle tissue, Nervous tissue
Organ	Two or more tissue types working together	Heart, Liver, Kidney
Organ System	Group of organs with related functions	Cardiovascular, Digestive
Organism	All systems together forming a living being	Human being

Chemical → Cellular → Tissue → Organ → Organ System → Organism

Example: *Atoms (H,C,O,N) → form a Cardiac Muscle Cell → many cells form Cardiac Muscle Tissue → tissue + nerve + vessels form the Heart (organ) → Heart + vessels form the Cardiovascular System → all systems together form the Human Being.*

1.2 Body Cavities

In easy words: Cavities are hollow spaces inside the body, like rooms in a house, that hold and protect our organs.



Picture 1.2 — The two main body cavities and what each one holds

Dorsal Cavity

- **Cranial cavity** — encloses the brain
- **Spinal (vertebral) cavity** — encloses the spinal cord

Ventral Cavity

- **Thoracic cavity** — heart, lungs, trachea
- **Abdominopelvic cavity** — digestive organs, kidneys, bladder, reproductive organs

Abdominopelvic 9 regions: Right Hypochondriac | Epigastric | Left Hypochondriac | Right Lumbar | Umbilical | Left Lumbar | Right Iliac | Hypogastric | Left Iliac

4 Quadrants: RUQ (liver, gallbladder) | LUQ (stomach, spleen) | RLQ (appendix, cecum) | LLQ (sigmoid colon)

Body Cavities

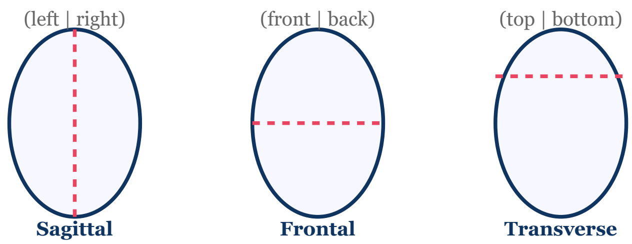
Dorsal Cavity ↳ Cranial (brain) | Spinal (spinal cord)

Ventral Cavity ↳ Thoracic (heart, lungs) | Abdominopelvic (stomach, kidneys, bladder)

Example: *A patient with appendicitis reports pain in the RLQ (Right Lower Quadrant) — appendix and cecum are located there.*

1.3 Body Planes (Anatomical Planes)

> **In easy words:** A "plane" is an imaginary flat cut through the body, used to describe where something is, like cutting an apple to see inside it.



Picture 1.3 — The three imaginary cuts (planes) used to describe body position

Plane	Also Called	Division
Sagittal	Lateral plane	Left and right halves; Midsagittal = equal halves
Frontal	Coronal plane	Anterior (front) and posterior (back)
Transverse	Horizontal / Axial plane	Superior (upper) and inferior (lower)

1.4 Directional Terms

Term	Meaning	Term	Meaning
Superior / Cranial	Toward the head	Inferior / Caudal	Toward the feet
Anterior / Ventral	Front of body	Posterior / Dorsal	Back of body
Medial	Toward midline	Lateral	Away from midline
Proximal	Closer to trunk	Distal	Farther from trunk
Superficial	Near body surface	Deep	Away from surface

1.5 Basic Physical Examination

1. Inspection
Visual assessment — skin, posture, breathing

2. Palpation
Feeling with hands — texture, tenderness, mass

3. Percussion
Tapping — dull (solid/fluid), resonant (air)

4. Auscultation
Listening via stethoscope — heart, lungs, bowel

Normal Vital Signs: Temperature 98.6°F / 37°C | Pulse 60–100 bpm | RR 12–20/min | BP 120/80 mmHg | SpO_2 95–100%

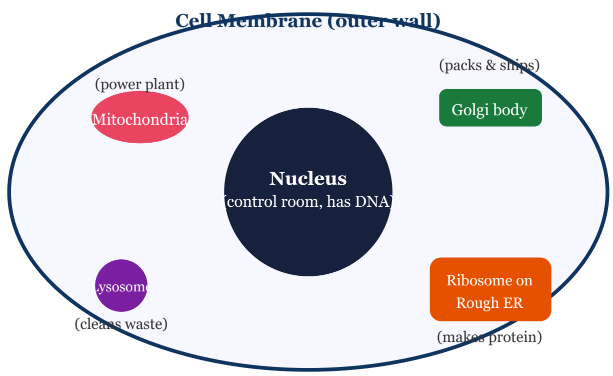
Inspection → Palpation → Percussion → Auscultation

Example: *Examining a patient with suspected pneumonia — look at breathing effort (inspection) → feel chest wall for tenderness (palpation) → tap chest, hear dull sound over consolidation (percussion) → listen with stethoscope for crackles (auscultation).*

Cell & Cell Division

2.1 Cell Structure — Key Organelles

In easy words: A cell is like a small living factory. Each part inside it (organelle) does one job — together they keep the cell alive and working.



Picture 2.1 — A simple labelled cell, showing where each part sits and what its job is

Organelle	Function
Cell membrane	Selective barrier; controls entry/exit
Nucleus	Control centre; contains DNA
Mitochondria	ATP production — powerhouse of the cell
Ribosome	Protein synthesis
Rough ER	Protein processing
Smooth ER	Lipid synthesis, detoxification
Golgi apparatus	Packages and ships proteins — post office of cell
Lysosome	Digests waste — suicide bag
Centrioles	Spindle formation during cell division

2.2 Cell Division

In easy words: Cells split into new cells in two ways — Mitosis makes 2 copy cells for growth and healing; Meiosis makes 4 special cells used to make babies.

Mitosis

- 2 identical daughter cells (diploid, 2n=46)
- Purpose: Growth, repair
- Phases: PMAT

Meiosis

- 4 genetically unique cells (haploid, n=23)
- Purpose: Gamete production
- Two rounds: Meiosis I and II

Mitosis Phase	Key Events
Prophase	Chromosomes condense; nuclear envelope breaks; spindle forms
Metaphase	Chromosomes align at equator (metaphase plate)
Anaphase	Sister chromatids separate to opposite poles
Telophase + Cytokinesis	Nuclear envelopes reform; cytoplasm divides into 2 cells

Prophase → Metaphase → Anaphase → Telophase → Cytokinesis

Example (PMAT): *A skin cell repairing a cut — chromosomes condense (P) → line up at the centre (M) → chromatids pull apart to opposite poles (A) → two new nuclei form and the cell splits (T) → 2 identical skin cells result, healing the wound.*

Clinical relevance: Uncontrolled mitosis → Cancer. Meiotic non-disjunction → Down syndrome (Trisomy 21), Turner (45,X), Klinefelter (47,XXY).

MODULE 2

Introduction to Medical Terminology & Word Formation

2.1 What is Medical Terminology?

Medical terminology is the special set of words doctors and nurses use to talk about the body, diseases, and treatments. Most of these words come from old **Greek and Latin** languages, so that doctors everywhere in the world understand the same word in the same way.

> **In easy words:** Think of medical words as Lego blocks — small pieces (prefix, root, suffix) snap together to build one big word.

2.2 Word Formation — Components

Component	Definition	Position	Example
Root Word	Core meaning	Middle	cardi (heart), hepat (liver)
Prefix	Modifies root; adds detail	Beginning	tachy- (fast), brady- (slow)
Suffix	Indicates condition/procedure	End	-itis (inflammation), -ectomy (removal)
Combining vowel	Links parts (usually 'o')	Between parts	cardiology (cardi-o-logy)

> **Rule:** Combining vowel used between root words or before consonant-beginning suffixes. Dropped before vowel-beginning suffixes. Example: gastr/o + itis = gastritis (dropped).

Prefix + Root + Combining Vowel + Suffix = Medical Term

Example 1: *peri- (around) + cardi (heart) + -itis (inflammation) =* ***Pericarditis*** *= inflammation around the heart.*
Example 2: *gastr/o (stomach) + enter/o (intestine) + -logy (study of) =* ***Gastroenterology*** *= study of the stomach and intestines.*

2.3 Greek & Latin Prepositional Prefixes

Prefix	Meaning	Example	Prefix	Meaning	Example
anti-	Against	Antibiotic	hypo-	Below normal	Hypoglycemia
peri-	Around	Pericardium	hyper-	Above normal	Hypertension
epi-	Upon, over	Epidermis	endo-	Within	Endoscopy
intra-	Within	Intravenous	sub-	Under	Subcutaneous
supra-	Above	Suprarenal	trans-	Across	Transdermal
pre-	Before	Prenatal	post-	After	Postoperative
inter-	Between	Intercostal	exo-	Outside	Exocrine

2.4 Key Prefixes — Number, Color, State

Prefix	Meaning	Example
mono-/uni-	One	Mononuclear
bi-/di-	Two	Bilateral
poly-/multi-	Many	Polyneuropathy
micro-	Small	Microcephaly
macro-/mega-	Large	Megacolon
a-/an-	Without	Apnea, Anemia
dys-	Difficult/painful	Dyspnea
brady-	Slow	Bradycardia
tachy-	Fast	Tachycardia
hemi-	Half	Hemiplegia

Color Prefix	Color	Example
erythr/o-	Red	Erythrocyte (RBC)
leuk/o-	White	Leukocyte (WBC)
cyan/o-	Blue	Cyanosis
melan/o-	Black/dark	Melanoma
xanth/o-	Yellow	Xanthoma
alb/o-	White	Albinism

2.5 Key Suffixes

Suffix	Meaning	Example
-algia/-dynia	Pain	Neuralgia
-itis	Inflammation	Hepatitis
-osis	Abnormal condition	Fibrosis
-emia	Blood condition	Anaemia
-pathy	Disease	Neuropathy
-rrhea	Flow/discharge	Diarrhoea
-plegia	Paralysis	Hemiplegia
-megaly	Enlargement	Hepatomegaly
-stenosis	Narrowing	Aortic stenosis

Suffix	Meaning	Example
-ectomy	Surgical removal	Appendectomy
-otomy	Surgical incision	Laparotomy
-ostomy	Surgical opening	Colostomy
-plasty	Surgical repair	Angioplasty
-scopy	Visual exam	Colonoscopy
-graphy	Recording process	Radiography
-gram	Record/image	ECG
-logy	Study of	Cardiology
-centesis	Surgical puncture	Amniocentesis

2.6 Common Root Words — Body Systems

Root	Meaning
cardi/o	Heart
angi/o	Blood vessel
hem/o	Blood
hepat/o	Liver
gastr/o	Stomach
nephr/o	Kidney
pulmon/o	Lung
neur/o	Nerve
encephal/o	Brain
dermat/o	Skin

Root	Meaning
oste/o	Bone
arthr/o	Joint
my/o	Muscle
hyster/o	Uterus
rhin/o	Nose
ophthalm/o	Eye
ot/o	Ear
adren/o	Adrenal gland
cyst/o	Bladder
col/o	Colon

MODULE 3

Human System Diseases & Procedures

♥ Cardiovascular System

In easy words: This is the heart and blood vessel system. Most problems here happen when blood flow to the heart is blocked or reduced.

Disease	Definition & Key Features
IHD	Ischaemic Heart Disease — reduced coronary blood flow; umbrella term (includes angina, MI)
Angina Pectoris	Chest pain due to insufficient myocardial O_2; stable (exertion) vs. unstable (rest)
Heart Valve Disease	Valvular stenosis (narrowing) or regurgitation (leaking) — mitral, aortic, tricuspid, pulmonary
CHD	Congenital Heart Disease — structural defect at birth (VSD, ASD, PDA, Tetralogy of Fallot)
Anaemia	Reduced Hb/RBC → low O_2 carrying capacity; Iron deficiency (most common), pernicious, aplastic
Thalassaemia	Inherited defect in globin chain synthesis → chronic haemolytic anaemia; alpha & beta types
Haemophilia	Inherited clotting factor deficiency; Type A (Factor VIII), Type B (Factor IX); X-linked recessive

Cardiovascular Procedures:

- **Angioplasty (PTCA)** — balloon catheter inflates in blocked coronary artery; stent placed
- **CABG** — bypass blocked coronary arteries using saphenous vein or IMA graft; open-heart surgery
- **CPR** — 30 chest compressions : 2 breaths; 100–120/min; restores circulation in cardiac arrest
- **Defibrillation** — electric shock to restore rhythm in VF/VT; AED available for public use
- **Pacemaker** — electrical device implanted for bradycardia; regulates slow heart rate
- **Heart Transplantation** — donor heart replacement for end-stage heart failure; lifelong immunosuppression

Coronary Artery Blockage → Reduced O_2 to Myocardium → Ischaemia (Angina) → Prolonged Blockage → Myocardial Infarction

Example: *A 55-year-old smoker gets chest pain on climbing stairs that resolves with rest — this is* ***stable angina*** *(exertional, reduced flow). If the same pain occurs at rest and lasts >20 minutes, it signals plaque rupture and possible* ***MI*** *— managed with angioplasty (PTCA) and stenting to reopen the artery.*

🍽 Digestive System

In easy words: This system breaks down food. Problems here usually involve acid, stones, or inflammation in the stomach, liver, or gallbladder.

Disease	Definition & Key Features
Peptic Ulcer	Open sore in stomach/duodenum lining; H. pylori or NSAIDs; burning epigastric pain
GERD	Gastro-Oesophageal Reflux Disease; backflow of acid; heartburn, regurgitation; weak LES
Dyspepsia	Indigestion; epigastric pain/bloating/nausea; functional or organic cause
Jaundice	Excess bilirubin → yellow skin/eyes; pre-hepatic / hepatic / post-hepatic classification
Gallstone (Cholelithiasis)	Hardened deposits in gallbladder; RUQ biliary colic; can cause cholecystitis
Hepatitis	Liver inflammation; viral (A/B/C/D/E), alcoholic, autoimmune; jaundice, elevated liver enzymes

Digestive Procedures:

- **Appendectomy** — removal of appendix; acute appendicitis; open or laparoscopic
- **Whipple Procedure** — pancreaticoduodenectomy; head of pancreas + duodenum + gallbladder; pancreatic cancer
- **Cholecystectomy** — gallbladder removal; mostly laparoscopic; for symptomatic gallstones
- **Hiatal Hernia Repair** — correction of stomach herniation through diaphragm
- **Nephrectomy** — kidney removal; cancer, trauma, donor transplant
- **Nissen Fundoplication** — wraps stomach fundus around lower oesophagus; strengthens LES; treats severe GERD

Weak LES → Acid Reflux into Oesophagus → Heartburn / Regurgitation → Chronic Irritation → Oesophagitis / Barrett's

Example: *A patient who feels burning behind the breastbone after eating spicy food and lying down at night has classic **GERD** due to a weak lower oesophageal sphincter (LES); persistent cases may need a **Nissen Fundoplication** to tighten the LES.*

🧠 Nervous System

In easy words: This is the brain, spinal cord, and nerves. Problems here affect movement, feeling, memory, or thinking.

Disease	Definition & Key Features
Stroke (CVA)	Sudden brain infarction; ischaemic (clot, 80%) or haemorrhagic; FAST = Face, Arm, Speech, Time
Quadriplegia	Paralysis of all 4 limbs; cervical spinal cord injury (C1–C7)
Bell's Palsy	Unilateral facial nerve (CN VII) paralysis; drooping of one side of face; usually reversible
Parkinson's Disease	Dopamine depletion; TRAP = Tremor, Rigidity, Akinesia, Postural instability
Alzheimer's Disease	Most common dementia; amyloid plaques + tau tangles; progressive memory loss; no cure

Nervous System Procedures:

- **Awake Brain Surgery (Awake Craniotomy)** — patient conscious; real-time monitoring near speech/motor cortex
- **Brain Aneurysm Surgery** — Clipping (surgical) or Coiling (endovascular); prevents subarachnoid haemorrhage
- **Epilepsy Surgery** — temporal lobectomy or corpus callosotomy; drug-resistant epilepsy
- **Locomotor Training** — treadmill + body-weight support; promotes neuroplasticity in SCI
- **Trigeminal Neuralgia Surgery** — Microvascular decompression (MVD); relieves pressure on CN V
- **Deep Brain Stimulation (DBS)** — electrode implant in subthalamic nucleus; treats Parkinson's, tremor

💧 Urinary System

In easy words: This system makes and removes urine. The kidneys filter waste from blood; problems here affect how well that filtering works.

Disease	Key Features
Dialysis	Artificial waste removal; haemodialysis (machine) vs. peritoneal (abdomen); used in CKD/ESRD
Nephritis	Kidney inflammation; glomerulonephritis; haematuria, proteinuria, oedema, hypertension
BPH	Benign Prostatic Hyperplasia; men >50; obstructs urethra; frequency, urgency, nocturia
Hydronephrosis	Kidney swelling from obstructed urine flow; stones, stricture, BPH; risk of renal damage

🧬 Endocrine System

In easy words: This system makes hormones (body messenger chemicals). Problems happen when a gland makes too much or too little hormone.

Disease	Key Features
Diabetes Mellitus	Type 1 (insulin deficiency) / Type 2 (insulin resistance); polydipsia, polyuria, polyphagia
Diabetic Foot	Neuropathy + poor circulation → ulcers, infection; high amputation risk
Gangrene	Tissue death; dry (vascular), wet (infected), gas (Clostridium); major diabetic complication
Hyposecretion diseases	Hypothyroidism, Addison's disease, Hypopituitarism, Hypoparathyroidism
Hypersecretion diseases	Hyperthyroidism/Graves', Cushing's (excess cortisol), Acromegaly (excess GH adult), Gigantism (excess GH child)

🔬 Reproductive System

In easy words: This system is about having babies. Problems here can affect periods, fertility, or the reproductive organs.

Disease	Key Features
Infertility	Failure to conceive after 12 months; female (ovulatory, tubal) or male (sperm) factors
Endometriosis	Endometrial tissue outside uterus; pelvic pain, dysmenorrhoea, infertility
PCOS	Polycystic Ovary Syndrome; hormonal; irregular periods, excess androgens, insulin resistance
PCOD	Polycystic Ovarian Disease; structural (multiple cysts); milder than PCOS
Uterine Fibroids	Benign smooth muscle tumours (leiomyomas); heavy periods, pelvic pressure; rarely cancerous
Syphilis	Treponema pallidum STI; stages: primary chancre → secondary rash → latent → tertiary; treated with penicillin
Prostatitis	Prostate inflammation; bacterial or non-bacterial; pelvic pain, dysuria
Erectile Dysfunction	Inability to maintain erection; vascular/neurological/hormonal/psychological; PDE5 inhibitors
Epididymitis	Epididymis inflammation; bacterial STI or UTI; scrotal pain/swelling; antibiotics

Reproductive Procedures:

- **Hysterectomy** — uterus removal; total/partial/radical; for fibroids, endometriosis, cancer, prolapse
- **Laparotomy** — abdominal cavity incision; exploratory surgery, ovarian cysts, obstetric emergencies

🫁 Respiratory System

In easy words: This is the breathing system — lungs and airways. Problems here make it hard to breathe properly.

Disease	Key Features
Asthma	Reversible bronchospasm; allergens/exercise triggers; wheeze, cough, dyspnoea; salbutamol + ICS
Pneumonia	Alveolar infection; S. pneumoniae (most common); fever, cough, pleuritic pain; CXR consolidation
Tuberculosis (TB)	Mycobacterium tuberculosis; cough, haemoptysis, night sweats; RIPE therapy × 6 months
COPD	Progressive airflow obstruction; smoking main cause; chronic bronchitis + emphysema; $FEV_1/FVC < 0.70$
Emphysema	Alveolar destruction; loss of elastic recoil; barrel chest; component of COPD; smoking

Key Distinctions: PCOS (hormonal syndrome) vs. PCOD (structural, milder) | Quadriplegia (cervical) vs. Paraplegia (thoracic+) | Haemodialysis (3×/week machine) vs. Peritoneal (daily, home) | Dry gangrene (vascular, no infection) vs. Wet (infected, spreads rapidly)

MODULE 4

Prescription Reading & Latin Abbreviations

4.1 Parts of a Prescription

In easy words: A prescription is a note from the doctor telling the pharmacist and patient which medicine to give, how much, and how often.

Part	Description
Superscription	Rx symbol + patient details (name, age, date)
Inscription	Drug name and strength
Subscription	Dispensing instructions to pharmacist (quantity)
Signa (Sig)	Patient directions — dose, route, frequency, duration
Refill	Number of refills allowed
Prescriber	Signature, reg. number, stamp

Superscription → Inscription → Subscription → Signa → Refill → Prescriber Sign

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Rx
Tab. Paracetamol 500mg
Sig: 1 tab PO TDS × 5 days, PC
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Example (reading the Sig above): *"1 tab PO TDS × 5 days, PC" means take **1 tablet, by mouth (PO), three times a day (TDS)**, for **5 days, after meals (PC)**.*

4.2 Frequency Abbreviations

Abbreviation	Latin	Meaning	Abbreviation	Latin	Meaning
OD / QD	Omni die	Once daily	HS	Hora somni	At bedtime
BD / BID	Bis in die	Twice a day	SOS / PRN	Si opus sit	If needed
TDS / TID	Ter in die	Three times a day	STAT	Statim	Immediately
QID / QDS	Quater in die	Four times a day	NPO / NBM	Nil per os	Nothing by mouth
AC	Ante cibum	Before meals	PC	Post cibum	After meals
CC	Cum cibum	With meals	WNL	—	Within Normal Limits

4.3 Route of Administration

Abbrev.	Meaning	Abbrev.	Meaning
PO	By mouth (per os)	IV	Intravenous
IM	Intramuscular	SC / SQ	Subcutaneous
SL	Sublingual	INH	Inhaled
PR	Per rectum	TOP	Topical
OD (eye)	Right eye	OS (eye)	Left eye
OU	Both eyes	ID	Intradermal

4.4 Dosage Forms

Abbreviation	Latin	Form	Abbreviation	Latin	Form
Tab	Tabella	Tablet	Inj	Injectio	Injection
Cap	Capsula	Capsule	Oint/Ung	Unguentum	Ointment
Syr	Syrupus	Syrup	Gtt	Guttae	Drops
Supp	Suppositorium	Suppository	Susp	Suspensio	Suspension

4.5 Clinical Abbreviations (Commonly Used)

Abbrev.	Full Form
Dx	Diagnosis
Hx	History
Sx	Symptoms
Tx	Treatment
k/c/o	Known case of
h/o	History of
c/o	Complaining of
R/O	Rule Out
NAD	No Abnormality Detected
NKDA	No Known Drug Allergies

Abbrev.	Full Form
CBC	Complete Blood Count
LFT	Liver Function Test
KFT/RFT	Kidney/Renal Function Test
FBS	Fasting Blood Sugar
ECG	Electrocardiogram
USG	Ultrasonography
CXR	Chest X-Ray
OPD	Outpatient Department
ICU	Intensive Care Unit
MLC	Medico-Legal Case

MODULE 5

Diagnostic Procedures

5.1 Fundamental Imaging Modalities

In easy words: "Imaging" means taking pictures inside the body without cutting it open, so doctors can see bones, organs, and problems clearly.

Modality	Principle	Radiation	Clinical Use
X-Ray	X-rays absorbed by dense tissue (bone), pass through soft tissue; 2D shadow image	Yes (low)	Fractures, CXR (pneumonia, TB), joints
USG	High-frequency sound waves reflect off tissues; echoes → image	None	Abdomen, obstetrics, thyroid, soft tissue; safe in pregnancy
CT Scan	Multiple X-ray beams rotate; computer reconstructs cross-sectional images	Yes (moderate-high)	Head trauma, stroke, tumours, internal injuries
MRI	Magnetic field + radiofrequency pulses; hydrogen proton realignment signals → image; no radiation	None	Brain, spine, joints, soft tissue; superior to CT for soft tissue
PET Scan	FDG radiotracer → cancer cells absorb more glucose → positron emission → gamma rays detected	Yes (radiotracer)	Cancer staging, Alzheimer's, cardiac viability; combined with CT = PET-CT
Biopsy	Tissue sample removal for pathological examination	None	Definitive cancer diagnosis; types: excisional, incisional, core needle, FNAC
FNAC	Fine needle (22–25G) aspirates cells for cytology	None	Thyroid nodules, breast lumps, lymph nodes; minimally invasive, outpatient

Suspected Bone Injury → X-Ray | Suspected Soft Tissue/Tumour → MRI/CT | Suspected Cancer Spread → PET-CT | Need Tissue Confirmation → Biopsy/FNAC

*Example: A patient with a wrist fracture after a fall first gets an **X-Ray** to confirm the broken bone; a patient with a suspicious breast lump gets **USG** first, then an **FNAC** to sample the cells and confirm whether it is cancerous.*

❤ Cardiology Department

Procedure	Description & Clinical Use
ECG	Electrocardiogram — records heart electrical activity; 12-lead standard; detects MI (STEMI = ST elevation), arrhythmias, heart blocks, hypertrophy; first-line cardiac test
Echocardiography	Cardiac ultrasound; assesses chambers, valves, ejection fraction, wall motion; types: 2D, 3D, Doppler, TOE; for heart failure, valve disease, CHD
Coronary Angiography	Contrast dye via catheter (femoral/radial) into coronary arteries; X-ray imaging; gold standard for CAD diagnosis; done in Cath Lab
Doppler USG	Measures blood flow direction and speed; colour Doppler maps flow; detects DVT, carotid stenosis, peripheral arterial disease
64-Slice CT	Non-invasive coronary CT angiography; detects hard and soft plaques; alternative to invasive angiography for low-moderate risk patients
Cardiac Catheterization	Catheter into heart chambers; measures pressures + O_2 levels; combined with angiography; for hemodynamic assessment + PCI interventions
Holter Monitor	Portable ECG worn 24–48 hours; detects intermittent arrhythmias (paroxysmal AF, SVT) missed on standard ECG

Gastroenterology Department

Procedure	Description & Clinical Use
Colonoscopy	Camera through rectum; visualizes entire colon + terminal ileum; CRC screening, IBD, polyps, biopsy + polypectomy
ERCP	Endoscopic Retrograde Cholangiopancreatography — endoscope to duodenum; contrast into bile/pancreatic ducts; treats bile duct stones (sphincterotomy), stenting; diagnostic + therapeutic
EGD (Upper GI Endoscopy)	Esophagogastroduodenoscopy — scope through mouth; visualizes oesophagus, stomach, duodenum; peptic ulcers, GERD, gastric cancer, H. pylori biopsy
Sigmoidoscopy	Short scope examining sigmoid colon + rectum; rectal bleeding, IBD; faster than colonoscopy; less preparation required
Esophageal Manometry	Pressure catheter in oesophagus; measures LES pressure + contractions; diagnoses achalasia, GERD, oesophageal spasm
Capsule Endoscopy	Swallowed pill-camera; images entire GI tract including small intestine; for obscure GI bleeding, Crohn's, small bowel tumours
MRCP	Magnetic Resonance Cholangiopancreatography — non-invasive MRI of biliary + pancreatic ducts; detects stones, strictures; preferred over diagnostic ERCP

🔬 Reproductive Department

Procedure	Description & Clinical Use
Colposcopy	Magnified examination of cervix after abnormal Pap smear; identifies dysplastic lesions; directed biopsy; cervical cancer screening
HSG	Hysterosalpingography — X-ray of uterus + fallopian tubes with contrast; assesses tubal patency; infertility workup
Scrotal USG	Ultrasound of testes, epididymis; detects testicular torsion, epididymitis, varicocele, testicular cancer; first-line for acute scrotal pain

🫁 Pulmonology Department

Procedure	Description & Clinical Use
Nuclear Lung Scanning (V/Q Scan)	Inhaled + IV radiotracer; gamma camera images; V/Q mismatch = pulmonary embolism; used when CTPA contraindicated (contrast allergy, pregnancy)
Pulmonary Artery Angiography	Catheter into pulmonary artery; contrast injection; X-ray imaging of pulmonary vasculature; gold standard for PE (now replaced by CTPA); essential pre-op for CTEPH surgery

🦴 Orthopaedic Department

Procedure	Description & Clinical Use
Bone Densitometry (DEXA)	Dual-energy X-ray measures Bone Mineral Density (BMD); T-score: Normal ≥ -1.0 | Osteopenia: -1.0 to -2.5 | Osteoporosis ≤ -2.5; post-menopausal women screening
Arthroscopy	Arthroscope inserted into joint through small incision; visualizes cartilage, ligaments, menisci; diagnostic + therapeutic (ACL repair, meniscectomy); most common: knee

Quick Reference — Modality Comparison:

Modality	Radiation	Invasive	Best For
X-Ray	Low	No	Bones, chest
USG	None	No	Abdomen, obstetrics
CT Scan	Moderate-high	No	Trauma, tumours, head
MRI	None	No	Brain, spine, soft tissue
PET Scan	Radiotracer	Injection only	Cancer, metabolism
Biopsy/FNAC	None	Yes (needle)	Tissue/cancer diagnosis
Angiography	Yes	Yes (catheter)	Coronary/pulmonary vessels